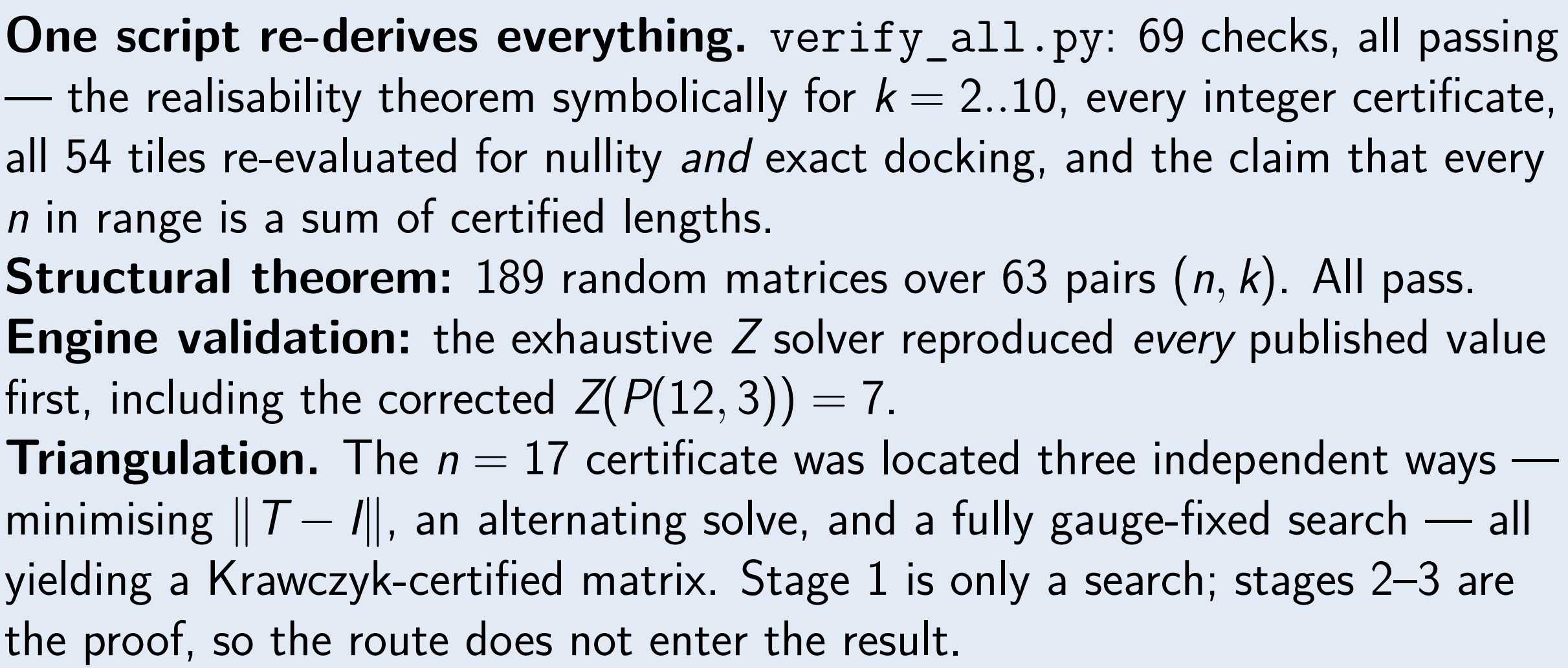


VERIFICATION



Why a failed search proves nothing here. *Lemma.* Every symmetric matrix on the pattern is gauge-equivalent to one in the slice we search. So a failure inside it measures effort, not impossibility — and this bit: $P(18, 3)$ and the tile $\ell = 21$ were both reported as failures and both succeeded once the restart budget was raised. *Working rule:* quote the residual and the budget, never just “failed”.

THE CERTIFICATIONS

k	settles	L	tiles	$\max_\alpha$
2	$n \geq 9, n \neq 10$	—	closed form	—
3	$n \geq 17$	21	23	2.1×10^{-10}
4	$n \geq 31$	31	31	3.8×10^{-10}

Each tile is an independent Krawczyk contraction: its own unknown count, its own $\|Y\|_\infty$, its own residual. Nothing is copied between them. At $k = 4$, $\ell = 61$: 1425 unknowns, 1220 equations of which 1175 are structural and 45 $\binom{10}{2}$ dependent, high-precision residual 4.7×10^{-61} .

5. A certificate spectrum: exact zeros at the ceiling
Spectrum of $\text{Adj} P(24, 5)$: nullity 12

Legend: 12 zero eigenvalues = $2k + 2$

CONCLUSIONS

1. The method has an **exact ceiling** of $2k + 2$, coinciding with the best known upper bound: either it settles Z completely or nothing does.
2. **Proved:** $Z(P(n, 2))$ and $Z(P(n, 3))$ **determined for every n** ; $Z(P(n, 4)) = 10$ for every $n \geq 30$. **No divisibility conditions.** The $k = 3$ sequence is 6, 6, 6, 8, 7, 7, 8, 8, ... — non-monotone, which is how the published proof went wrong.
3. **Proved:** the congruence conditions of the cyclotomic layer were *forced*, not a shortfall — one class, at most one prime. That converts the earlier results into a demarcation.
4. **Proved:** $c^2 = 0$ iff the symbol factors as $(s + a)(L_k + t)$; over $\mathbb{Q}$, Niven's theorem caps the degenerate symbols at 25 per k .
5. **Corrected (my own), among others:** the claim this construction “never exceeds 6”; a false unconditional cover bound ($Z(P(8, 2)) = 5$); the threshold $(2k + 2)^2/5$; $N(k) = O(k)$; and an eigenvalue search whose negative results I **withdrew** after it reported nullity 7 at $P(20, 3)$ where 8 is attained.

FUTURE WORK

Honest limits. $k \geq 5$ is *not* settled: tiles exist at $k = 5$ from $\ell \approx 54$, but $[54, 108)$ needs 54 of them and we have 19. Four values at $k = 4$ remain, $n \in \{24, 25, 28, 29\}$. The $k \geq 6$ congruence-class values are numerical, a weaker standard than the rest. The rank law is verified over a range, not proved.

What we cannot diagnose. We first searched $k = 5$ over $41 \leq \ell \leq 45$ — the parameter count's bound — and found nothing, then found tiles starting near 54. The count is a lower bound, *not* a predictor of where tiles begin (21 vs 16 at $k = 3$; ≈ 54 vs 41 at $k = 5$), and we cannot explain the gap.

Next. Close $[54, 108]$ for $k = 5$ (mechanical); prove the rank law; decide whether $M < Z$ strictly at $n = 10$, $k = 2$.

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- Next.** Close $[54, 108]$ for $k = 5$ (mechanical); prove the rank law; decide whether $M < Z$ strictly at $n = 10$, $k = 2$.

KEY REFERENCES

[1] S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli, *Computing the zero forcing number for generalized Petersen graphs*, J. Algebra Combin. Discrete Struct. Appl. **7**(2) (2020) 183–193.

[2] A. Krishnan, *A correction to the zero forcing number of the generalized Petersen graphs $P(n, 3)$* , arXiv:2607.19412 (2026).

[3] AIM Minimum Rank Special Graphs Work Group, *Zero forcing sets and the minimum rank of graphs*, Linear Algebra Appl. **428** (2008) 1628–1648.